MIL-P-48077 (MU)
26 June 1972
SUPERSEDING
(See 6.8)

MILITARY SPECIFICATION

PRIMER: M104,

PARTS FOR

AND

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts, and the loading, assembling and packing for one type of primer designated as Primer: M104.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-403	-	Plastic Molding Material Polytetra Fluoroethylene (TFE-Fluorocarbon)

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes-(ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

C-8864967	-	Primer, M104
F-9258182	-	Box, Set-up, Packing Ammunition, Inner, for Small Explosive Components (2 SHTS)
C-9258183	-	Box, Set-up, Packing, Ammunition, Outer, for Small Explosive Components (2 SHTS)
C-9258184	-	Box, Packing, Ammunition, for Small Explosive Components (2 SHTS)

PUBLICATIONS

ARMY

ET-8864967	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

Source: https://assist.dla.mil -- Downloaded: 20260-26T03:55Z
Check the source to verify that this is the current version before use.

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primer.-The primer shall comply with all requirements specified on Drawing (Dwg.) C-8864967, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Moisture content of priming mixture.-The moisture content of priming mixture at the time of loading shall be less than 0.50 percent when tested as specified in 4.4.1.

3.4 Moisture content of lead azide.-The moisture content of the lead azide increment at the time of loading shall be less than 0.30 percent when tested as specified in 4.4.2.

3.5 Functioning.-The primer shall function on the first attempt under the impact of a two (2)-ounce ball falling a distance of four (4) inches, maximum, when tested as specified in 4.4.3.

3.6 Output.-The explosive output shall indent the specified disc to a depth of 0.125 inch, minimum, when tested as specified in 4.4.3.

3.7 Adherence of primers to packaging.-The primers shall not stick to the release paper or spacer in the inner set-up box when inspected as specified in 4.4.4.

3.8 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, surface defects, dirt, grease, rust, and other foreign matter. The cleaning method shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

3.9 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.6)

3.10 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, with the exception that a statement of findings on the moisture content of each explosive shall be furnished with each lot submitted for acceptance.
d. Number of items in the lot.
e. Date submitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of one-hundred and fifty (150) primers.

All parts and completed assemblies shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the same to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All primers shall be subjected by the Government to the examinations specified in 4.3.2.3. In addition, the following tests shall be performed:

4.2.2.1 Functioning (See 3.5).-One-hundred (100) primers shall be subjected to this test. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.2.2.2 Output (See 3.6).-Fifty (50) primers shall be subjected to this test. The test shall be performed as specified in 4:4.3 using equipment in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.2.3 Rejection.-If any primer fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any primer in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Parts of one lot interfix number from one manufacturer.
b. Priming mixture from one batch or blending of batches which results in homogeneity of all priming mixture used in loading the entire primer lot.
c. Lead azide from one batch or blending of batches which results in homogeneity of all lead azide used in loading the entire primer lot.
d. Lead styphnate from one batch or blending of batches which results in homogeneity of all lead styphnate used in loading the entire primer lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, primer (See Dwg. C-8864968).

Categories		Defects	Method of Inspection	Code Number (See 6.2)
Critical:	None defined			
Major:		AQL 0.65 Percent		
	101.	Wall thickness, minimum (min.)	Gage	01001
	102.	Concentricity of hole diameter with outside diameter	Gage	01002
	103.	Diameter of hole	Gage	01003
Minor:		AQL 1.50 Percent		
	201.	Total length	Gage	01004
	202.	Finish improper	Visual	01005
	203.	Evidence of poor workmanship (See 3.8)	Visual	01006

4.3.2.2 Primer (prior to loading) (See Dwg. C-8864967).

Categories		Defects	Method of Inspection	Code Number
Critical:	None defined			
Major:	None defined			
Minor:		AQL 0.65 Percent		
	201.	Foreign matter on interior (See 3.8)	Visual	02001

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.3.2.3 Primer (See Dwg. C-8864967).

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Explosive on exterior	Visual	03001
2.	Explosive exposed (disc punctured, cut or torn)	Visual	03002
3.	Cup cut, split or cracked	Visual	03003
Major:	AQL 0.65 Percent		
101.	Total length (prior to sealing)	Gage	03004
102.	Outside diameter, maximum	Gage	03005
103.	Color improper at ends	Visual	03006
Minor:	AQL 2.50 Percent		
201.	Paint on ends not dry	Manual	03007
202.	Distortion	Visual	03008
203.	Disc wrinkled or folded	Visual	03009
204.	Crimp not a full 360 degrees	Visual	03010
205.	Crack at crimp	Visual	03011
206.	Evidence of poor workmanship (See 3.8)	Visual	03012

4.3.2.4 Box, set-up, inner (after loading and prior to closing slide cover) (See Dwg. F-9258182).

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Cushion-filler assembly missing	Visual	04001
2.	Stiffener missing from slide cover	Visual	04002
3.	Primers improperly positioned	Visual	04003
Major:	AQL 0.40 Percent		
101.	Release paper missing or improperly assembled	Visual	04005
Minor:	AQL 1.00 Percent		
201.	Marking missing, incomplete or illegible	Visual	04006
202.	Evidence of poor workmanship (See 3.8)	Visual	04007

4.3.2.5 Box, set-up, inner (after closing slide cover) (See Dwg. F-9258182).

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Slide cover assembled reversed	Visual	05001

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Major: None defined

Minor: None defined

4.3.2.6 Box, set-up, outer (with inner set-up boxes assembled and prior to assembling top and sealing strip) (See Dwg. C-9258183).

Categories	Defects	Method of Inspection	Code Number
Critical:			
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	06001
2.	Black surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	06002
3.	All stiffeners not facing one direction	Visual	06003
Major:	AQL 0.40 Percent		
101.	Box damaged to the extent that contents are exposed	Visual	06004
Minor:	AQL 1.50 Percent		
201.	Marking missing, incomplete or illegible, including incorrect color code	Visual	06005
202.	Evidence of poor workmanship (See 3.8)	Visual	06006
203.	Label missing or badly wrinkled	Visual	06007

4.3.2.7 Box, packing (prior to assembling cover and strapping) (See Dwg. C-9258184).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Waterproofness of bag destroyed(separation of heat seal, or rip, tear, or puncture)	Visual	07001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	07002

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code Number
Minor:	AQL 1.00 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	07003
202.	Marking misleading or unidentifiable, including incorrect color code	Visual	07004

4.3.2.8 Box, sealed wooden packing (See Dwg. C-9258184).

Categories	Defects	Method of Inspection	Code Number
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Hardware missing, broken or loose	Visual-Manual	08001
102.	Strapping missing, broken or loose	Visual-Manual	08002
103.	Contents exposed	Visual	08003
104.	Board broken or split	Visual	08004
105.	DoD symbol misleading or unidentifiable	Visual	08005
Minor:	AQL 2.50 Percent		
201.	Hardware improperly engaged	Visual-Manual	08006
202.	Contents loose	Manual	08007
203.	Strapping improperly assembled	Visual	08008
204.	Marking misleading or unidentifiable, including incorrect color code	Visual	08009
205.	Car seal missing, unsealed, or improperly positioned	Visual	08010

4.3.3 Testing.

4.3.3.1 Moisture content of priming mixture (See 3.3) Major defect (Code Number 09001).-At least one six (6)-gram sample of priming mixture shall be selected for the moisture content determination from that quantity to be used in each day's production of primers. Sample priming mixture shall be selected at the site of, and immediately before loading. If the moisture content exceeds the specified requirement, the priming mixture shall not be used until corrective action has been taken as verified by repeating this test. Primers already loaded with non-conforming priming mixture shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.2 Moisture content of lead azide (See 3.4) Major defect (Code Number 10001).-At least one six (6)-gram sample of lead azide shall be selected for the moisture content determination from that quantity of lead azide to be used in each day's production of primers. The sample shall be selected at the site of, and immediately before loading. If the moisture

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

content exceeds the specified requirement, the lead azide shall not be used until corrective action has been taken as verified by repeating this test. Primers already loaded with non-conforming lead azide shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.3 Functioning and output (See 3.5 and 3.6) Major defect (Code Number 11001-functioning test only).-The sampling plans for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. If any primer fails to comply with the requirements specified in 3.5 and 3.6, it shall be classed defective. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4. (See 6.6).

4.3.3.4 Output (See 3.6) Major defect (Code Number 12001).-Primer sensitivity shall be ascertained for each lot of primers. A sample of at least 60 primers shall be selected from each lot for this test. The staircase method shall be used to ascertain the height of 50 percent functioning (h) and standard deviation (s). The quality and mixing of ingredients shall be checked if the h and s exceed the values of 1.35 and 0.50 inches, respectively. All results shall be recorded. (See 6.6).

4.3.3.5 Adherence of primers to packaging (See 3.7) Major defect (Code Number 13001).-A sample of twenty-five (25) inner set-up boxes from each lot shall be selected for this inspection. Not more than one (1) inner set-up box shall be selected from each outer set-up box opened and, in addition, the samples shall be selected a minimum of forty-eight (48) hours after assembly of the inner set-up boxes into the outer set-up boxes. If a total of twenty-one (21) or more primers from the sample stick to either the release paper or inner set-up box, the lot shall be rejected. The inspection shall be performed as specified in 4.4.4.

4.3.4 Inspection equipment. Equipment Tabulation (ET) number 8864967 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DoD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.4 Test methods and procedures.

4.4.1 Moisture content of priming mixture.-Transfer an accurately weighed portion of approximately one gram of the priming mixture to a tared aluminum weighing dish provided with a teflon cover while both the dish and cover are on the balance pan. Place the dish and contents, with the cover removed, in an oven and dry at 60° ± 5°C. for 2 hours. The cover shall be placed in a desiccator during the drying period. After drying, transfer the dish and contents quickly to a desiccator and cool to room temperature. Replace the plastic cover and weigh. Calculate the loss in weight of the dish, cover and contents to percent moisture in the sample.

4.4.2 Moisture content of lead azide.-A tared dried aluminum dish, 2 inches in diameter, shall be covered with a sheet of teflon (See L-P-403), 2-1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams shall be transferred to the moisture dish while both the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95°C plus or minus (±) 5°C. for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{(A - B) \times 100}{A}$$

where

A = the weight of the specimen before drying.
B = the weight of the specimen after drying.

4.4.3 Functioning and output.-The primers shall be mounted in the test fixture shown on the Equipment List. A weight of two ounces, maximum, shall be dropped from a height of four inches, maximum, on the firing pin. Satisfactory functioning and output will be evidenced by explosion of the primers and by the size of the depression in the disc as specified in 3.5 and 3.6.

4.4.4 Adherence of primers to packaging.-Visual inspection shall be performed to determine whether any primers are sticking to the inner set-up box. A primer "transfer block" (See 6.9) shall be used for this determination. The "transfer block" shall be placed on the top of the spacer in the inner set-up box and the box then inverted so the primers fall out of the spacer and into the holes in the transfer block. Visual inspection shall then be performed to determine whether any primers remain in the spacer. Any primer remaining in the box because of being "cocked" shall not be considered a defect. Equipment for this inspection shall be satisfactory to the Contracting Officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level B (See 6.7) and Level C (See 6.1).-The primers shall be preserved and packaged in accordance with Dwgs. F-9258182 and C-9258183.

5.2 Packing.

5.2.1 Level B (See 6.7) and Level C (See 6.1).-The primers, preserved and packaged as specified in 5.1.1, shall be packed in accordance with Dwg. C-9258184.

5.3 Marking.-Marking shall be in accordance with Dwgs. F-9258182, C-9258183 and C-9258184 and Code of Federal Regulations, Title 49, Parts 171-179.

5.4 Shipping.-When shipments of more than one lot are made as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.
d. Level of packaging and packing required. (See 6.7)

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design - mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection wtih non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Analysis of composition.-The following procedures are provided for use in the event that quantitative analyses are desired for informational purposes:

6.4.1 Barium Nitrate Content.-Transfer an accurately weighed portion of approximately 0.5 gram of the dried sample (dried as described in 4.3.3.2) to a tared medium-porosity, fritted glass crucible of 30 milliliter (ml) capacity. Record the weight of crucible and sample as Weight A. Add 3 ml. of distilled water, saturated with lead azide at a temperature of 50° ± 2°C. Swirl for 30 seconds, then apply suction. Repeat this procedure two times, making a total of three 3-ml. extractions. Using an eye dropper, transfer to the crucible enough

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

alcohol to thoroughly wet the sample and wash the crucible walls. Aspirate until the odor of alcohol is no longer detectable and dry the crucible in an oven for 30 minutes at 60° ± 5°C. Cool in a desiccator and weigh. Record this weight as Weight B. Reserve the crucible and residue for the basic lead styphnate determination. Correct for the percentage of basic lead styphnate in the water extract as follows: Transfer the filtrate and washings quantitatively to a 100-ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark. Using a calibrated pipette, transfer a 10-ml aliquot of this solution to a second 100-ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell (See 6.4.6) having a width of approximately 1 centimeter (cm), and determine its optical density at a wave length of 410 millimicrons. Use as reference a similar cell containing the same proportion of water saturated with lead azide and alcohol as the sample solution. Since the cell which holds the lead styphnate solution is not identical with the reference cell, correct the optical density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with a solution of water and alcohol, and measure, at a wave length of 410 millimicrons, the optical density of the cell which originally contained the lead styphnate solution. Calculate the percentage of lead styphnate in the filtrate on a dry basis as follows:

Basic lead styphnate found in barium nitrate filtrate,

$$\text{Percent } P_b = \frac{4.48\ (A-B)}{WD}$$

where:

A = Optical density of normal lead styphnate solution.
B = Optical density of water and alcohol solution.
D = Width of corex cell, cm.
W = Weight of dried sample.

Calculate the percentage of barium nitrate in the sample as follows:

$$\text{Barium nitrate, percent} = \frac{100\ (A-B)}{W} - P_b$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

where:

W = Weight of the dried sample.
A = Weight of crucible and sample.
B = Weight of the crucible and sample after extractions with water saturated with lead azide.
P_b = Percentage of basic lead styphnate found in barium nitrate filtrate.

6.4.2 Lead styphnate.-The residue obtained in the determination of barium nitrate shall be extracted with six-ml portions of ammonium acetate solution at about 25°C., (hot ammonium acetate shall not be used as it dissolves antimony sulfide to some extent). Agitate each portion for about 30 seconds allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water, saturated with antimony sulfide until the washings coming through the crucible appear colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water, and diluted to the mark. Using a calibrated pipette, a 2-ml aliquot of this solution shall be transferred to a 50 ml calibrated volumetric flask, and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately one centimeter (cm) and its optical density (See 6.4.6) determined at a wave length of 410 millimicrons. As a reference, a similar cell containing ammonium acetate solution of the same concentration as the solution of the specimen shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the optical density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. This shall be done by filling both cells with ammonium acetate solution and measuring at a wave length of 410 millimicrons, the optical density of the cell which originally contained the lead styphnate solution. The percentage of basic lead styphnate in the specimen shall be calculated as follows:

$$\text{Percentage basic lead styphnate} = \frac{28\ (A-B)}{WD} + P_b$$

where:

A = Optical density of basic lead styphnate.
B = Optical density of acetate solution.
D = Width of corex cell, cm.
W = Weight of dried specimen.
P_b = Percent basic lead styphnate found in barium nitrate filtrate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

6.4.3 Tetracene.-The wet residue obtained in the determination of basic lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60° ± 5°C. for 30 minutes, cooled in a desiccator, and weighed. The dried residue in the crucible shall be transferred to a 125-ml beaker with a stream of water, 25 ml of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into same crucible and the residue washed three times with boiling water and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60° ± 5° C. for 30 minutes, cooled in a desiccator, and weighed. The percentage of tetracene in the specimen shall be calculated as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W}$$

where:

C = Weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = Weight of crucible and residue after treating with hot water.
W = Weight of the dried specimen.

6.4.4 Antimony sulfide content.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the specimen as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W}$$

where:

D = Weight of crucible and specimen after treating with hot water.
E = Weight of crucible.
W = Weight of dried specimen.

6.4.5 Lead azide content.-The lead azide content in the specimen shall be determined by subtracting from 100 the combined percentage of barium nitrate, basic or normal lead styphnate, tetracene and antimony sulfide.

6.4.6 Optical density determination.-A Quartz Ultraviolet Spectrophotometer, described in the Journal of American Optical Society, Vol. 30, Page 683 (1941), was found to be satisfactory in determining the optical density of the styphnate solution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

6.5 Check test for possible deterioration (not applicable to new production).-The following test is designed to prevent the use of primers that have deteriorated. This check test shall be performed on primers from each lot which have been stored for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of this check test. The contracting officer should arrange to have the contractor reimbursed for any expense incurred in his performance of the check test for possible deterioration.

6.5.1 Functioning.-The sampling plan for the test shall be in accordance with Military Standard MIL-STD-105, using Code letter J and an AQL of 0.65 percent.

6.5.2 Output.-The sampling plan for this test shall be in accordance with Military Standard MIL-STD-105, using Code letter H and an AQL of 1.00 percent.

6.6 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.7 Level B packaging and packing.-In the event the expected storage time of the primers is in excess of 180 days, the barrier bag shall be used as specified on the applicable drawing. Expected storage time shall be determined by the procuring agency.

6.8 Supersession data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-93 (Rev. 3), dated 30 April 1965.

6.9 Transfer block.-The "transfer block" shall be a block of suitable material (wood, plastic, etc.) with holes to match the holes in the spacer in the inner set-up box, and so designed to permit the primers to fall out of the box, when it is inverted, into the matching holes in the transfer block.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A011

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters, Picatinny Arsenal
Dover, New Jersey 07801

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-QA-A
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.